iDesk

Enterprise IT Helpdesk System

Blueprint Teknis Lengkap — End to End

Versi Dokumen	1.0
Tanggal	24 Februari 2026
Status	Production Active
Total Modul Backend	28 Modules
Total Fitur Frontend	17 Feature Areas
Total Entitas Database	40+ Entities

BLUEPRINT APLIKASI WEB iDesk

Enterprise IT Helpdesk System â€" Dokumentasi Teknis Lengkap

| Field | Detail | |---|---| | **Nama Aplikasi** | iDesk â€" Enterprise IT Helpdesk System | | **Versi Dokumen** | 1.0 | | **Tanggal** | 24 Februari 2026 | | **Status** | Production Active | |

Frontend URL | http://localhost:4050 | | **Backend API** | http://localhost:5050 | | **Swagger Docs** | http://localhost:5050/api |

---

1. EXECUTIVE SUMMARY

iDesk adalah sistem helpdesk dan ticketing enterprise full-stack (NestJS + React). Mengelola seluruh siklus hidup permintaan IT support dengan integrasi multi-channel dan otomasi. **Tiga Portal:** | Portal | Pengguna | Fungsi | |--------|----------|--------| | Admin/Agent Portal | ADMIN & AGENT | Kelola tiket, laporan, konfigurasi | | Manager Portal | MANAGER | Monitoring, laporan eksekutif | | Client Portal | USER | Buat tiket, pantau, knowledge base | **Statistik:** 28 Backend Modules, 17 Frontend Feature Areas, 44+ UI Components, 40+ DB Entities

---

2. ARSITEKTUR SISTEM

2.1 Stack Overview

- **Backend**: NestJS 10 + TypeORM + PostgreSQL â€" Port 5050
- **Frontend**: React 18 + Vite + TailwindCSS â€" Port 4050
- **Real-time**: Socket.IO WebSocket bidirectional
- **Queue/Cache**: Redis (Bull queues, optional)
- **Bot**: Telegram via Telegraf (polling/webhook)
- **Container**: Docker + Docker Compose

2.2 Frontend Routing

```
PUBLIC: /login â†' Login page /feedback/:token â†' Survey page /unauthorized â†' Access denied ADMIN/AGENT PORTAL (/): /dashboard â†' Analytics dashboard /kanban â†' Kanban board /tickets/list â†' Ticket list + filter /tickets/:id â†' Ticket detail + chat /tickets/create â†' New ticket /agents â†' User management [ADMIN] /reports â†' Reports PDF/Excel /sla â†' SLA config [ADMIN] /renewal â†' Contract management /kb, /kb/manage â†' Knowledge base /settings â†' System settings /notifications â†' Notification center /automation â†' Workflow rules [ADMIN] /workloads â†' Agent workload [ADMIN] /audit-logs â†' Audit trail [ADMIN] /system-health â†' Health dashboard [ADMIN] /zoom-calendar â†' Zoom booking /zoom-
```

```
settings â†’ Zoom config [ADMIN] MANAGER PORTAL (/manager): /manager/dashboard â†’ Manager overview /manager/workloads â†’ Team workloads /manager/tickets â†’ All tickets view /manager/reports â†’ Executive reports /manager/kb â†’ Knowledge base /manager/zoom-calendar â†’ Zoom calendar /manager/renewal â†’ Contract renewal CLIENT PORTAL (/client): /client/my-tickets â†’ My tickets /client/create â†’ Submit ticket /client/tickets/:id â†’ Ticket detail /client/notifications â†’ Notifications /client/zoom-calendar â†’ Zoom calendar /client/kb â†’ Knowledge base /client/profile â†’ User profile `
```

---

3. TECHNOLOGY STACK

Backend

```
| Teknologi | Kegunaan | |-----------|----------| | NestJS 10 | Framework modular | | TypeORM | ORM PostgreSQL | | Socket.IO | Real-time WebSocket | | Passport JWT | Auth + RBAC | | Telegraf | Telegram Bot | | Bull | Redis job queues | | PDFKit + ExcelJS | Report generation | | Nodemailer + Handlebars | Email SMTP + template | | @nestjs/schedule | Cron jobs | | @nestjs/event-emitter | Event-driven automation | | Multer | File upload | | Swagger | API documentation | | Helmet + Throttler | Security + rate limiting | | bcrypt | Password hashing | | Google APIs | Spreadsheet sync |
```

Frontend

```
| Teknologi | Kegunaan | |-----------|----------| | React 18 | UI library | | Vite | Build tool | | TailwindCSS 3 | Styling | | Radix UI | Accessible components | | TanStack Query | Server state | | Zustand | Client state | | React Router DOM v6 | Routing | | Framer Motion | Animations | | Recharts | Charts | | Socket.IO Client | WebSocket | | React Hook Form + Zod | Form + validation | | Tiptap | Rich text editor | | Sonner | Toast notifications |
```

Infrastructure

```
| Komponen | Teknologi | |----------|-----------| | Database | PostgreSQL 15+ | | Cache/Queue | Redis 7+ | | Container | Docker Compose | | Storage | Local ./uploads |
```

---

4. AUTENTIKASI & OTORISASI

4.1 Credential & Token

- JWT Bearer Token (stateless)

  - Token disimpan di localStorage auth-storage

- bcrypt password hashing
- Role-based expiry: Admin/Agent = 3h, User = 1h

4.2 Roles

| Role | Deskripsi | Default Password (seed) | |------|-----------|---------| | ADMIN | Administrator penuh | admin123 | | AGENT | IT Support staff | agent123 | | MANAGER | IT Manager | - | | USER | End user/karyawan | user123 |

4.3 Guards & Middleware

| Guard | Fungsi | |-------|--------| | JwtAuthGuard | Verifikasi JWT token | | RolesGuard | Cek role pada endpoint | | PageAccessGuard | Cek izin per halaman | | FeatureAccessGuard | Cek izin per fitur | | CustomThrottlerGuard | Rate limiting global |

4.4 Security Features

- Rate limiting: 100 req/60s per IP
- IP Whitelist: blokir IP tidak diizinkan
- Page lockout: max 10 denial â†’ lockout 15 menit
- Helmet security headers
- Input validation via class-validator pada DTO
- Magic bytes check untuk upload file
- CORS protection

---

5. CORE TICKETING SYSTEM

5.1 Tiket â€” Field Lengkap

| Field | Tipe | Keterangan | |-------|------|------------| | id | UUID | Primary key | | ticketNumber | String unique | Auto-generated nomor tiket | | title | String | Judul masalah | | description | Text | Deskripsi lengkap | | category | String | Kategori (GENERAL, dll) | | device | String | Nama perangkat | | software | String | Software terkait | | status | Enum | TODO/IN_PROGRESS/WAITING_VENDOR/RESOLVED/CANCELLED | | priority | String | LOW/MEDIUM/HIGH/CRITICAL | | ticketType | Enum | SERVICE/ICT_BUDGET/LOST_ITEM/ACCESS_REQUEST/HARDWARE_INSTALLATION/ORACLE_REQUEST | | source | Enum | WEB/TELEGRAM/EMAIL | | userId | UUID | Pembuat | | assignedToId | UUID | Agent ditugaskan | | siteId | UUID | Site/cabang | | slaTarget | Date | Deadline SLA | | firstResponseTarget | Date | Target respons pertama | | isOverdue | Boolean | Sudah terlambat | | isFirstResponseBreached | Boolean | Respons pertama terlambat | | criticalReason | Text | Alasan prioritas CRITICAL | | isHardwareInstallation | Boolean | Flag instalasi hardware | | scheduledDate | Date | Jadwal instalasi | | scheduledTime | String | Jam jadwal HH:mm | | hardwareType | String | PC/IP_PHONE/PRINTER | | version | Int | Optimistic locking |

5.2 Status Flow

`TODO â†’ IN_PROGRESS â†” WAITING_VENDOR â†’ RESOLVED â†’ CANCELLED`

5.3 Fitur Ticketing

| Fitur | Deskripsi | |-------|-----------| | Multi-source | Buat tiket via Web, Telegram, Email | | Rich text messages | Editor WYSIWYG + attachment | | Internal notes | Pesan private agent-only | | @mention | Tag agent dalam percakapan | | File attachment | Gambar, PDF, dokumen | | Canned responses | Template jawaban cepat | | Ticket templates | Form pre-filled | | Bulk operations | Mass update status/priority/assignee | | Ticket surveys | Rating kepuasan post-resolve | | Custom attributes | Field tambahan per kategori | | SLA tracking | Countdown + breach alert | | Time entries | Log waktu pengerjaan | | Hardware scheduling | Jadwal instalasi + reminder otomatis | | Kanban view | Drag-drop per kolom status | | List view | Tabel dengan sort + filter + search |

5.4 Ticket Message Entity

| Field | Keterangan | |-------|------------| | id | UUID | | ticketId | Referensi tiket | | senderId | User pengirim | | content | Isi pesan (rich text) | | isInternal | Flag internal note | | attachments | Array file attachment | | readBy

| Array user yang sudah baca | | createdAt | Timestamp |

---

6. MANAJEMEN PENGGUNA

6.1 User Entity

| Field | Keterangan | |-------|----------| | id | UUID primary key | | email | Unique, login identifier | | password | bcrypt hashed | | fullName | Nama lengkap | | role | ADMIN/AGENT/MANAGER/USER | | employeeId | ID karyawan | | jobTitle | Jabatan | | phoneNumber | Nomor telepon | | departmentId | FK ke Department | | avatarUrl | URL foto profil | | telegramId | Linked Telegram account | | telegramChatId | Chat ID Telegram | | telegramNotifications | Toggle notif Telegram | | isActive | Status aktif | | lastActiveAt | Timestamp aktivitas terakhir | | siteId | FK ke Site (multi-site) | | appliedPresetId | Permission preset aktif | | appraisalPoints | Poin kinerja agent | | createdAt / updatedAt | Timestamps |

6.2 Fitur User Management

| Fitur | Deskripsi | |-------|----------| | CRUD Users | Create, Read, Update, Delete user | | CSV Import | Bulk import user dari file CSV | | Avatar Upload | Upload foto profil | | Department Management | Kelola departemen | | Telegram Linking | Hubungkan akun Telegram | | Role Management | Atur role & izin | | Permission Preset | Template izin per tipe user | | Activity Indicator | Status online terakhir | | Soft Delete | Deaktivasi user tanpa hapus data | | Password Reset | Admin reset password |

---

7. SLA MANAGEMENT

7.1 Konsep SLA

SLA (Service Level Agreement) mendefinisikan tenggat waktu penyelesaian tiket berdasarkan prioritas.

7.2 SLA Config Entity

| Field | Keterangan | |-------|----------| | id | UUID | | priority |

LOW/MEDIUM/HIGH/CRITICAL | | responseTimeMinutes | Target waktu respons pertama | | resolutionTimeMinutes | Target waktu resolusi | | isActive | Status aktif |

7.3 Business Hours

| Field | Keterangan | |-------|------------| | dayOfWeek | 0-6 (Minggu-Sabtu) | | startTime | Jam mulai kerja (HH:mm) | | endTime | Jam selesai kerja (HH:mm) | | isWorkingDay | Hari kerja atau tidak | | holidays | Daftar hari libur nasional Indonesia (2025-2026) |

7.4 Cara Kerja SLA

`

- Tiket dibuat â†’ slaStartedAt = now
- Hitung slaTarget berdasarkan priority + business hours
- Cron job tiap 5 menit â†’ cek semua tiket aktif
- Jika slaTarget - now < threshold â†’ kirim SLA warning
- Jika now > slaTarget â†’ isOverdue = true + kirim breach alert
- Status WAITING_VENDOR â†’ clock di-pause (totalWaitingVendorMinutes++)
- SLA resume ketika status kembali ke IN_PROGRESS

`

7.5 SLA Alerts

- Warning email ke assignee & admin saat mendekati deadline
- Telegram notification untuk agen
- Dashboard indicator merah pada tiket overdue
- Auto-reassign jika dikonfigurasi di workflow rules

---

8. KNOWLEDGE BASE

8.1 Article Entity

| Field | Keterangan | |-------|------------| | id | UUID | | title | Judul artikel | | content | Isi artikel (rich text/HTML) | | summary | Ringkasan singkat | | category | Kategori artikel | | tags | Array tags | | authorId | Penulis | | visibility | PUBLIC/INTERNAL/PRIVATE | | isPublished | Status publikasi | | viewCount | Jumlah tampilan | | helpfulCount | Jumlah "artikel ini membantu" | |

`notHelpfulCount | Jumlah "tidak membantu" | | createdAt / updatedAt | Timestamps |`

8.2 Fitur Knowledge Base

```
| Fitur | Deskripsi | |-------|-----------| | Create/Edit Article | Editor WYSIWYG lengkap | | Categories & Tags | Organisasi konten | | Visibility Control | PUBLIC/INTERNAL/PRIVATE | | Full-text Search | Cari artikel berdasarkan kata kunci | | View Tracking | Hitung popularitas artikel | | Helpful/Not Helpful | Feedback rating | | Article Management | Daftar kelola untuk admin/agent | | Client Access | User dapat mengakses KB publik |
```

---

9. NOTIFICATION SYSTEM

9.1 Notification Channels

```
| Channel | Deskripsi | |---------|-----------| | In-App | Notifikasi dalam aplikasi (real-time via WebSocket) | | Email | SMTP email dengan template HTML Handlebars | | Telegram | Pesan langsung ke akun Telegram user | | Push | Browser push notification (PWA) |
```

9.2 Notification Entities

```
| Entity | Keterangan | |--------|------------| | Notification | Notifikasi per user | | NotificationPreference | Preferensi channel per user | | NotificationLog | Log pengiriman (success/fail) | | PushSubscription | Data push subscription browser |
```

9.3 Event yang Memicu Notifikasi

```
| Event | Penerima | |-------|---------| | Tiket baru dibuat | Assignee + Admin | | Tiket di-assign | Agent ditugaskan | | Status tiket berubah | Pembuat tiket | | Pesan baru di tiket | Semua partisipan | | SLA mendekati deadline | Assignee + Admin | | SLA breach | Assignee + Admin | | Tiket resolved | Pembuat tiket | | Renewal kontrak mendekati expired | Admin | | Zoom booking dikonfirmasi/dibatalkan | Peserta | | Hardware installation reminder H-1 & H-0 | User & Agent |
```

---

10. AUTOMATION ENGINE (WORKFLOW)

10.1 Konsep

Engine otomasi berbasis rule: **Trigger â†’ Conditions â†’ Actions**. Setiap rule mendengarkan event tertentu dan menjalankan aksi jika kondisi terpenuhi.

10.2 WorkflowRule Entity

Field	Keterangan
id	UUID
name	Nama rule
isActive	On/Off toggle
priority	Urutan eksekusi (angka lebih kecil = prioritas lebih tinggi)
trigger	JSON: { type, event }
conditions	JSON Array: [{ field, operator, value }]
actions	JSON Array: [{ type, params }]

10.3 Trigger Types (9 Jenis)

Trigger	Keterangan
TICKET_CREATED	Tiket baru dibuat
TICKET_STATUS_CHANGED	Status tiket berubah
TICKET_PRIORITY_CHANGED	Prioritas berubah
TICKET_ASSIGNED	Tiket di-assign ke agent
MESSAGE_RECEIVED	Pesan baru masuk
SLA_WARNING	SLA mendekati deadline
SLA_BREACHED	SLA sudah terlampaui
TICKET_IDLE	Tiket tidak ada aktivitas selama X jam
SCHEDULED	Berbasis jadwal/waktu

10.4 Condition Operators (14 Jenis)

EQUALS, NOT_EQUALS, CONTAINS, NOT_CONTAINS, STARTS_WITH, ENDS_WITH, IN, NOT_IN, GREATER_THAN, LESS_THAN, IS_NULL, IS_NOT_NULL, MATCHES_REGEX, NOT_MATCHES_REGEX

10.5 Action Types (7 Jenis)

Aksi	Keterangan
CHANGE_STATUS	Ubah status tiket
CHANGE_PRIORITY	Ubah prioritas tiket
ASSIGN_TO	Assign ke agent spesifik
SEND_NOTIFICATION	Kirim notifikasi ke user/group
SEND_EMAIL	Kirim email
SEND_WEBHOOK	HTTP callback ke URL eksternal
ADD_TAG	Tambah tag/label tiket

10.6 Auto-Assignment

Mode	Deskripsi
ROUND_ROBIN	Distribusi merata bergiliran antar agent
LEAST_BUSY	Assign ke agent dengan tiket aktif paling sedikit

10.7 Pre-built Templates

5 template bawaan tersedia:

- High Priority Escalation â€” eskalasi tiket HIGH/CRITICAL
- SLA Warning Notification â€” ingatkan sebelum breach
- Auto-assign New Tickets â€” round-robin assignment
- Close Resolved Tickets â€” auto-close setelah X hari resolved
- Routing by Category â€” arahkan tiket berdasarkan kategori

---

11. LAPORAN & ANALITIK

11.1 Jenis Laporan

| Laporan | Format | Keterangan | |---------|--------|------------| | Monthly Summary | PDF + Excel | Ringkasan tiket bulanan | | Agent Performance | PDF | Metrik per agent | | SLA Compliance | PDF | Tingkat kepatuhan SLA | | Ticket by Category | Excel | Distribusi per kategori | | Resolution Time | Excel | Rata-rata waktu penyelesaian | | Trend Analysis | Chart | Grafik tren tiket |

11.2 Dashboard Analytics (Real-time)

Widget di BentoDashboard:

- Total tiket (hari ini / bulan ini / all-time)
- Tiket by status (TODO, IN_PROGRESS, WAITING, RESOLVED, CANCELLED)
- Tiket overdue
- Average response time
- SLA compliance rate
- Agent performance comparison
- Ticket volume trend (chart line)
- Category distribution (pie chart)
- Top performing agents

11.3 Manager Dashboard

Khusus role MANAGER:

- Team workload overview
- Per-agent ticket count

- Resolution rate comparison
- SLA performance heatmap
- Reports download

---

12. INTEGRASI TELEGRAM BOT

12.1 Fitur Bot

| Fitur | Deskripsi | |-------|-----------| | Create ticket | Buat tiket baru via chat | | View ticket status | Cek status tiket aktif | | Receive notifications | Notifikasi real-time | | Two-way messaging | Balas tiket via Telegram | | Role-based menus | Menu berbeda per role | | Account linking | Hubungkan akun iDesk dengan Telegram |

12.2 Role-based Menu

| Role | Menu yang Tersedia | |------|-------------------| | USER | Buat tiket, lihat tiket saya, KB | | AGENT | Lihat tiket assigned, update status, kirim pesan | | ADMIN | Semua menu + statistik singkat |

12.3 Mode Operasi

- **Polling**: Untuk development (terus-menerus tanya API Telegram)
- **Webhook**: Untuk production (Telegram push ke server)

12.4 TelegramSession Entity

| Field | Keterangan | |-------|------------| | id | UUID | | telegramId | Telegram user ID | | chatId | Telegram chat ID | | userId | Linked iDesk user ID | | state | State machine percakapan | | context | Data konteks sementara | | lastActiveAt | Aktivitas terakhir |

---

13. CONTRACT RENEWAL (RENEWAL HUB)

13.1 RenewalContract Entity

| Field | Keterangan | |-------|------------| | id | UUID | | contractName | Nama

kontrak | | vendorName | Nama vendor | | contractNumber | Nomor kontrak | | startDate | Tanggal mulai | | expiryDate | Tanggal kadaluarsa | | value | Nilai kontrak | | currency | Mata uang | | category | Kategori (SOFTWARE/HARDWARE/SERVICE) | | status | ACTIVE/EXPIRING/EXPIRED/RENEWED | | documentUrl | URL file kontrak (PDF) | | notes | Catatan tambahan | | acknowledgedBy | User yang konfirmasi | | acknowledgedAt | Waktu konfirmasi |

13.2 Fitur Renewal

| Fitur | Deskripsi | |-------|-----------| | PDF Parsing | Auto-ekstrak data dari kontrak PDF | | Manual Entry | Input manual jika PDF tidak bisa diparse | | Expiry Alerts | Notifikasi 30/60/90 hari sebelum expired | | Acknowledgment | Tracking konfirmasi renewal | | Dashboard | Overview status semua kontrak | | VPN Access Tracking | Pantau akses VPN per user | | Google Sync | Sinkronisasi ke Google Spreadsheet |

---

14. ZOOM BOOKING CALENDAR

14.1 Entities

| Entity | Keterangan | |--------|------------| | ZoomAccount | Akun Zoom terdaftar (S2S OAuth) | | ZoomBooking | Reservasi meeting | | ZoomMeeting | Data meeting Zoom aktual | | ZoomParticipant | Peserta meeting | | ZoomSettings | Pengaturan global | | ZoomAuditLog | Log perubahan booking |

14.2 ZoomBooking Fields

| Field | Keterangan | |-------|------------| | title | Judul meeting | | description | Deskripsi | | bookingDate | Tanggal meeting | | startTime / endTime | Jam mulai & selesai (HH:mm) | | durationMinutes | Durasi dalam menit | | status | PENDING/CONFIRMED/CANCELLED | | bookedByUserId | User yang booking | | zoomAccountId | Akun Zoom yang digunakan | | cancellationReason | Alasan pembatalan | | participants | Daftar peserta |

14.3 Fitur Zoom Calendar

| Fitur | Deskripsi | |-------|-----------| | Visual Calendar | Tampilan kalender bulanan | | Slot Availability | Cek ketersediaan slot | | Booking | Reservasi meeting Zoom | | Cancellation | Batalkan dengan alasan | | participants | Tambah

`peserta | | Auto-create Zoom meeting | Otomatis buat meeting di Zoom | | Notifikasi | Email & in-app ke semua peserta | | Audit trail | Log semua perubahan | | Multi-account | Kelola beberapa akun Zoom | | All roles access | Admin/Agent/Manager/User bisa akses |`

---

15. SISTEM IZIN AKSES (PERMISSIONS)

15.1 Entities

`| Entity | Keterangan | |--------|------------| | FeatureDefinition | Definisi fitur yang dikontrol | | UserFeaturePermission | Izin per user per fitur | | PermissionPreset | Template izin yang bisa diterapkan ke banyak user |`

15.2 Cara Kerja

```
- Admin buat FeatureDefinition (e.g., "zoom_calendar")
- Admin buat PermissionPreset (e.g., "IT Staff Preset")
- Preset berisi konfigurasi izin per fitur
- Admin terapkan preset ke user/group
- Saat user akses halaman â†’ PageAccessGuard cek UserFeaturePermission
- Jika tidak ada permission â†’ redirect ke /unauthorized
```

15.3 Fitur yang Dikontrol Permission

- zoom_calendar
- knowledge_base
- reports
- renewal
- notifications
- settings
- Dan dapat ditambah fitur baru

---

16. GOOGLE SPREADSHEET SYNC

16.1 Entities

| Entity | Keterangan | |--------|------------| | SpreadsheetConfig | Konfigurasi spreadsheet tujuan | | SpreadsheetSheet | Sheet dalam spreadsheet | | SyncLog | Log sinkronisasi |

16.2 Fitur

| Fitur | Deskripsi | |-------|-----------| | Auto-sync | Sync data secara berkala ke Google Sheets | | Manual sync | Trigger sync manual | | Multiple sheets | Sinkronisasi ke beberapa sheet | | Sync log | Riwayat sync berhasil/gagal | | Kolom mapping | Peta kolom data ke spreadsheet | | Service Account auth | Auth via Google Service Account JSON |

16.3 Use Cases

- Export data tiket ke Google Sheets (monitoring)
- Renewal contract tracking di Sheets
- VPN access log di Sheets
- Custom data sync sesuai kebutuhan

---

17. MODUL TAMBAHAN

17.1 ICT Budget Request

Form permintaan anggaran untuk pembelian IT:

- Judul & deskripsi kebutuhan
- Estimasi biaya
- Justifikasi
- Status approval
- Integrasi dengan ticket type ICT_BUDGET

17.2 Lost Item Report

Laporan kehilangan barang milik perusahaan:

- Deskripsi barang
- Lokasi terakhir diketahui

- Tanggal kehilangan
- Status investigasi

17.3 Access Request

`Permintaan akses ke sistem/aplikasi/jaringan:`

- Jenis akses yang diminta (AccessType entity)
- Alasan kebutuhan akses
- Durasi akses
- Status approve/reject

17.4 VPN Access Tracking

`Pantau dan kelola akses VPN per user:`

- Request akses VPN
- Approval workflow
- Log akses
- Revoke akses

17.5 IP Whitelist Management

`Kelola IP yang diizinkan mengakses sistem:`

- CRUD IP address/range
- Enable/disable per IP
- Deskripsi/alasan whitelist

17.6 Agent Workload & Appraisal

`| Fitur | Deskripsi | |-------|-----------| | PriorityWeight | Bobot poin per prioritas tiket | | AgentDailyWorkload | Tracking beban kerja harian | | Appraisal points | Akumulasi poin kinerja per agent | | Workload dashboard | Visualisasi distribusi beban kerja |`

17.7 Synology Backup Integration

`| Fitur | Deskripsi | |-------|-----------| | BackupConfiguration | Konfigurasi backup ke Synology NAS | | BackupHistory | Riwayat backup berhasil/gagal | | Scheduled backup | Backup otomatis terjadwal | | Status monitoring | Monitor`

`status backup |`

17.8 Sound Notifications

`Kustomisasi suara notifikasi:`

- `NotificationSound entity`
- `Upload custom sound file`
- `Assign sound per event type`
- `Toggle per user`

17.9 Multi-Site Support

`| Fitur | Deskripsi | |-------|-----------| | Site entity | Kelola multiple cabang/lokasi | | Site isolation | Tiket & user ter-isolasi per site | | Site-based reporting | Laporan per site | | Cross-site admin | Admin bisa akses semua site |`

17.10 System Health

`| Fitur | Deskripsi | |-------|-----------| | Health check endpoint | GET /health | | Database connectivity | Cek koneksi PostgreSQL | | Redis connectivity | Cek koneksi Redis | | Memory usage | Monitor penggunaan RAM | | System info | Versi, uptime, environment |`

17.11 Audit Log

`| Field | Keterangan | |-------|------------| | id | UUID | | userId | User yang melakukan aksi | | action | Jenis aksi (CREATE/UPDATE/DELETE) | | entityType | Tipe entitas (Ticket, User, dll) | | entityId | ID entitas yang terpengaruh | | oldValues | Nilai sebelum perubahan (JSON) | | newValues | Nilai setelah perubahan (JSON) | | ipAddress | IP address user | | createdAt | Timestamp |`

---

18. SEARCH SYSTEM

18.1 Global Search

- `Unified search across: Tiket, Artikel KB, User`
- `Command palette (shortcut Ctrl+K / Cmd+K)`

- Real-time autocomplete
- Filter by type

18.2 Saved Search

| Field | Keterangan | |-------|------------| | id | UUID | | userId | Pemilik | | name | Nama pencarian | | query | String pencarian | | filters | JSON filter kriteria | | createdAt | Timestamp |

---

19. FILE UPLOAD

| Fitur | Deskripsi | |-------|-----------| | Upload endpoint | POST /uploads | | Supported types | Image (jpg/png/gif/webp), PDF, Office docs | | Size limit | Konfigurasi via env | | Magic bytes check | Validasi tipe file sebenarnya | | Static serving | GET /uploads/:filename | | Drag & drop UI | React Dropzone di frontend |

---

20. API ENDPOINTS UTAMA

Auth

| Method | Endpoint | Deskripsi | |--------|----------|-----------| | POST | /auth/login | Login + get JWT | | GET | /auth/profile | Data user aktif | | POST | /auth/refresh | Refresh token |

Tickets

| Method | Endpoint | Deskripsi | |--------|----------|-----------| | GET | /tickets | List tiket + filter | | POST | /tickets | Buat tiket baru | | GET | /tickets/:id | Detail tiket | | PATCH | /tickets/:id | Update tiket | | DELETE | /tickets/:id | Hapus tiket | | GET | /tickets/:id/messages | Pesan tiket | | POST | /tickets/:id/messages | Kirim pesan | | PATCH | /tickets/bulk | Bulk update |

Users

| Method | Endpoint | Deskripsi | |--------|----------|-----------| | GET | /users | List user | | POST | /users | Buat user | | PATCH | /users/:id | Update user | | DELETE | /users/:id | Hapus user | | POST | /users/import-csv | Import CSV | | GET

| /departments | List departemen |

Knowledge Base

| Method | Endpoint | Deskripsi | |--------|----------|-----------| | GET | /kb/articles | List artikel | | POST | /kb/articles | Buat artikel | | GET | /kb/articles/:id | Detail artikel | | PATCH | /kb/articles/:id | Edit artikel | | DELETE | /kb/articles/:id | Hapus artikel |

Notifications

| Method | Endpoint | Deskripsi | |--------|----------|-----------| | GET | /notifications | List notifikasi | | PATCH | /notifications/read-all | Tandai semua dibaca | | GET | /notifications/preferences | Preferensi notif | | PATCH | /notifications/preferences | Update preferensi |

Reports

| Method | Endpoint | Deskripsi | |--------|----------|-----------| | GET | /reports/monthly | Laporan bulanan | | GET | /reports/agent-performance | Kinerja agent | | GET | /reports/export/pdf | Export PDF | | GET | /reports/export/excel | Export Excel |

Zoom Booking

| Method | Endpoint | Deskripsi | |--------|----------|-----------| | GET | /zoom-booking/accounts | List akun Zoom | | GET | /zoom-booking/bookings | List booking | | POST | /zoom-booking/bookings | Buat booking | | PATCH | /zoom-booking/bookings/:id | Update booking | | DELETE | /zoom-booking/bookings/:id | Cancel booking |

Automation

| Method | Endpoint | Deskripsi | |--------|----------|-----------| | GET | /automation/rules | List workflow rules | | POST | /automation/rules | Buat rule | | PATCH | /automation/rules/:id | Update rule | | DELETE | /automation/rules/:id | Hapus rule |

---

21. REAL-TIME SYSTEM (WEBSOCKET)

21.1 Events yang Di-emit Backend â†’ Frontend

| Event | Payload | Keterangan | |-------|---------|------------| | ticket:created | Ticket data | Tiket baru dibuat | | ticket:updated | Ticket data | Tiket diupdate | | ticket:status-changed | { id, status } | Status berubah | | ticket:message | Message data | Pesan baru masuk | | notification:new | Notification | Notifikasi untuk user | | agent:online | { userId } | Agent aktif | | zoom:booking-updated | Booking data | Booking Zoom berubah |

21.2 Rooms & Namespaces

- User masuk room berdasarkan role dan ticket ID
- Notifikasi dikirim hanya ke room yang relevan
- Heartbeat connection management

---

22. KONFIGURASI ENVIRONMENT

`env

Database PostgreSQL

DB_HOST=localhost DB_PORT=5432 DB_USERNAME=postgres DB_PASSWORD=your-secure-password DB_DATABASE=idesk_db DB_POOL_MIN=2 DB_POOL_MAX=10

JWT Authentication

JWT_SECRET=generate-a-strong-64-char-secret JWT_EXPIRES_IN=60m

Telegram Bot

TELEGRAM_BOT_TOKEN=your-telegram-token TELEGRAM_USE_WEBHOOK=false # true for

```
production TELEGRAM_WEBHOOK_DOMAIN=https://api.yourapp.com
```

Redis (optional)

```
REDIS_ENABLED=false REDIS_HOST=localhost REDIS_PORT=6379 REDIS_PASSWORD=
```

Email SMTP

```
SMTP_HOST=smtp.ethereal.email SMTP_PORT=587 SMTP_SECURE=false SMTP_USER=your-smtp-user SMTP_PASS=your-smtp-pass SMTP_FROM='iDesk Support '
```

Zoom API (Server-to-Server OAuth)

```
ZOOM_ACCOUNT_ID=your-zoom-account-id ZOOM_CLIENT_ID=your-zoom-client-id ZOOM_CLIENT_SECRET=your-zoom-client-secret
```

Google API

```
GOOGLE_CREDENTIALS_PATH=path/to/service-account.json
```

Application

```
PORT=5050 NODE_ENV=development FRONTEND_URL=http://localhost:4050
```

`UPLOAD_DIR=./uploads`

Rate Limiting

`THROTTLE_TTL=60000 THROTTLE_LIMIT=100 PAGE_ACCESS_CACHE_TTL=300 PAGE_ACCESS_MAX_DENIALS=10 PAGE_ACCESS_LOCKOUT_MINUTES=15` `

---

23. INFRASTRUKTUR & DEPLOYMENT

23.1 Development Setup

` `bash`

1. Clone repository

`git clone https://github.com/HitterD/iDesk.git cd iDesk`

2. Setup environment

`cp .env.example .env`

Edit .env sesuai kebutuhan

3. Start database (Docker)

```
docker-compose -f docker-compose.db.yml up -d
```

Atau: deploy_database_docker.bat (Windows)

4. Install dependencies

`npm run install:all`

5. Run migrations

`cd apps/backend && npm run migration:run`

6. Seed database

`npm run seed`

7. Start all services

`npm start`

Atau: startup.bat (Windows)

`

23.2 Default Login Credentials

`| Role | Email | Password | |------|-------|----------| | Admin | admin@idesk.com | admin123 | | Agent | agent@idesk.com | agent123 | | User | user@idesk.com | user123 |`

23.3 Ports

`| Service | Port | |---------|------| | Frontend (Vite dev) | 4050 | | Backend (NestJS) | 5050 | | PostgreSQL | 5432 | | Redis | 6379 | | Swagger UI | 5050/api |`

23.4 Docker Services (docker-compose.yml)

`| Service | Image | Keterangan | |---------|-------|------------| | postgres | postgres:15 | Database utama | | redis | redis:7 | Cache + queue (optional) | | backend | Custom Dockerfile | NestJS app | | frontend | Custom Dockerfile | React app |`

23.5 Database Scripts

```bash

# Backup database

backup_db.bat

# Run migrations

npm run migration:run

# Generate migration

npm run migration:generate -- -n MigrationName

# Revert migration

npm run migration:revert
```

---

24. STRUKTUR KOMPONEN FRONTEND (44+ Components)

24.1 Layout Components

- BentoLayout `â€” Admin/Agent portal shell (sidebar + topbar)`
- ClientLayout `â€” Client portal shell`
- ManagerLayout `â€” Manager portal shell`
- Sidebar `â€” Navigation sidebar dengan collapsible`
- TopBar `â€” Header dengan search, notif, user menu`

24.2 Core UI Components

- Button `,` Input `,` Select `,` Checkbox `,` RadioGroup
- Dialog/Modal `,` Popover `,` Dropdown
- Card `,` Badge `,` Avatar

- Table, DataGrid
- Progress, Slider, Switch
- Tabs, Accordion
- Toast /Sonner notifications
- LoadingScreen, Skeleton loaders

24.3 Advanced Feature Components

- CommandPalette â€” Global search (Ctrl+K)
- RichTextEditor â€” Tiptap WYSIWYG
- VirtualizedList â€” Performance list untuk data besar
- ActivityFeed â€” Real-time activity stream

- CannedResponses `â€” Quick reply picker`
- ArticleSearchAutocomplete `â€” KB search inline`
- BulkActionsToolbar `â€” Mass operations UI`
- PDFPreviewModal `â€” Preview PDF in-browser`
- ErrorBoundary `+` FeatureErrorBoundary `â€” Error isolation`
- ScreenReaderAnnounce `â€” Accessibility`

24.4 Custom Hooks (19 hooks)

- useAuth `â€” Auth state & actions`
- useTickets `â€” Ticket CRUD + realtime`
- useNotifications `â€” Notification center`
- useSocket `â€” WebSocket management`

- usePermissions `â€” Permission checks`
- useDebounce `â€” Search input debounce`
- useLocalStorage` â€" Typed localStorage
- Dan lainnya...

---

25. KEAMANAN SISTEM

| Aspek | Implementasi | |-------|----------| | Password | bcrypt hashing dengan salt | | JWT | Signed tokens, role-based expiry | | Rate Limiting | 100 req/menit per IP | | IP Whitelist | Blokir IP tidak diizinkan | | Page Access | Guard per halaman + lockout | | File Upload | Magic bytes validation | | Headers | Helmet (X-XSS, HSTS, CSP) | | Input | class-validator pada semua DTO | | CORS | Konfigurasi origin whitelist | | SQL Injection | TypeORM query builder (parameterized) | | Slow Query | Log query > 1 detik | | DB Pool | Min 5, Max 20 connections |

---

26. RINGKASAN FITUR Aâ€"Z

| Fitur | Deskripsi | Role | |-------|-----------|------| | Access Request | Permintaan akses sistem/jaringan | All | | Agent Management | Kelola user agent & admin | ADMIN | | Appraisal Points | Poin kinerja agent otomatis | ADMIN | | Audit Trail | Log semua perubahan sistem | ADMIN | | Automation Rules | Workflow IF-THEN otomatis | ADMIN | | Backup Synology | Integrasi backup ke NAS Synology | ADMIN | | Bulk Operations | Mass update tiket | ADMIN/AGENT | | Business Hours | Konfigurasi jam kerja SLA | ADMIN | | Canned Responses | Template jawaban cepat | AGENT | | Contract Renewal | Manajemen kontrak IT dengan alert | ADMIN/AGENT | | CSV Import | Bulk import user | ADMIN | | Dashboard Analytics | Statistik real-time | ADMIN/AGENT | | Department Management | Struktur organisasi | ADMIN | | Events & Notifications | Notifikasi multi-

channel | All | | File Attachments | Upload dokumen ke tiket | All | | Google Sheets Sync | Sinkronisasi data ke Google Sheets | ADMIN | | Hardware Installation | Request + jadwal instalasi hardware | All | | Health Monitoring | Status sistem real-time | ADMIN | | ICT Budget Request | Pengajuan anggaran IT | All | | Internal Notes | Note private antar agent | AGENT/ADMIN | | IP Whitelist | Kelola izin akses per IP | ADMIN | | JWT Authentication | Login aman dengan token | All | | Kanban Board | Visual drag-drop ticket management | ADMIN/AGENT | | Knowledge Base | Artikel panduan & FAQ | All | | Lost Item Report | Laporan kehilangan barang | All | | Manager Dashboard | Monitoring khusus manajer | MANAGER | | Multi-site Support | Manajemen multi cabang | ADMIN | | Notification Center | Pusat semua notifikasi | All | | Oracle Access Request | Permintaan akses sistem Oracle | All | | Permission Presets | Template izin per tipe user | ADMIN | | PDF Reports | Generate laporan PDF | ADMIN/MANAGER | | Push Notifications | Browser push notification | All | | Rate Limiting | Proteksi brute-force | System | | Real-time Updates | WebSocket live updates | All | | Reports & Analytics | Laporan performa | ADMIN/MANAGER | | Role-based Access | Akses granular per peran | System | | Saved Replies | Simpan teks jawaban | AGENT/ADMIN | | Search Global | Cari tiket/artikel/user | All | | Settings | Preferensi sistem & profil | All | | SLA Management | Tracking & alert keterlambatan | System | | Sound Notifications | Kustomisasi suara notif | All | | Surveys/Feedback | Rating kepuasan setelah tiket selesai | All | | Telegram Bot | Buat & pantau tiket via Telegram | All | | Ticket Templates | Form pre-filled untuk tiket umum | ADMIN/AGENT | | Time Tracking | Log waktu pengerjaan tiket | AGENT | | User Management | CRUD pengguna sistem | ADMIN | | VPN Access Tracking | Kelola & pantau akses VPN | ADMIN/AGENT | | Workload Dashboard | Distribusi beban kerja agent | ADMIN/MANAGER | | WYSIWYG Editor | Editor teks kaya untuk tiket & KB | All | | Zoom Booking | Kalender booking meeting Zoom | All |